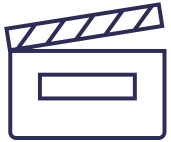
# Agents

# Introduction

# Agents

What is an Agent?

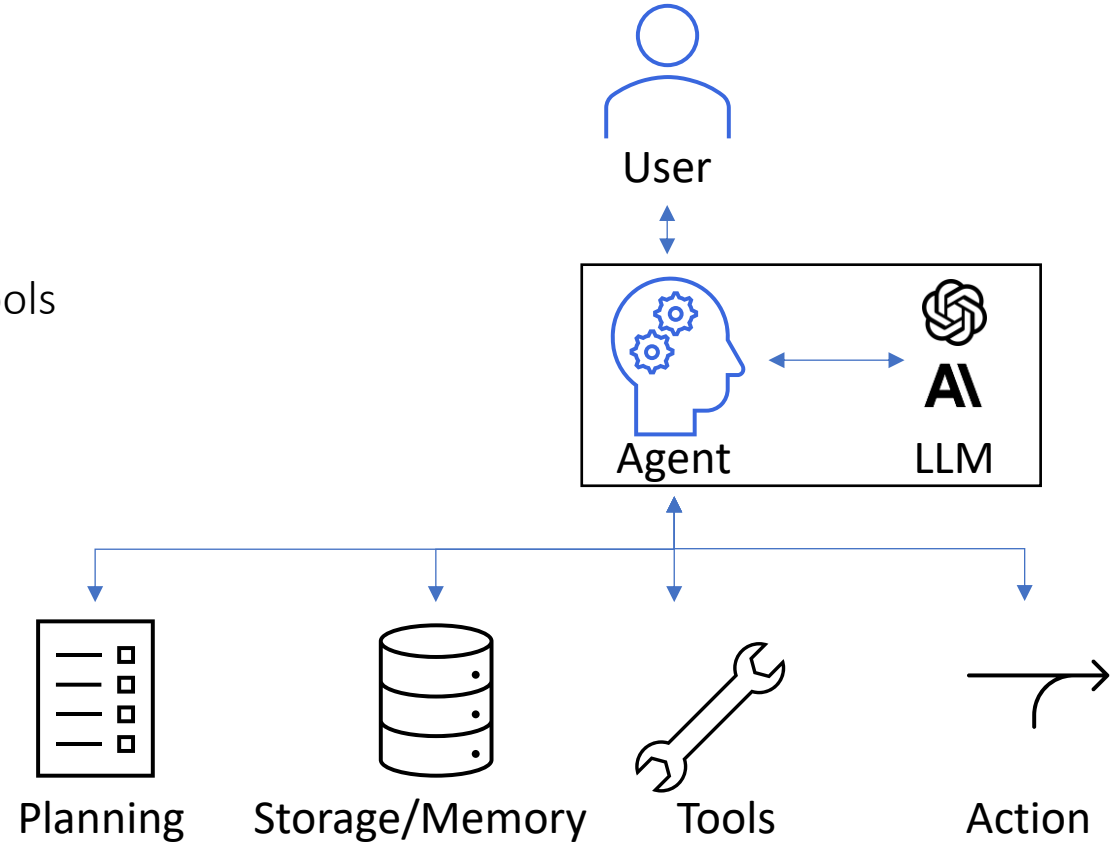
Perceives Environment → Takes Actions autonomously → To achieve a defined goal → Improve through learning



# Agents

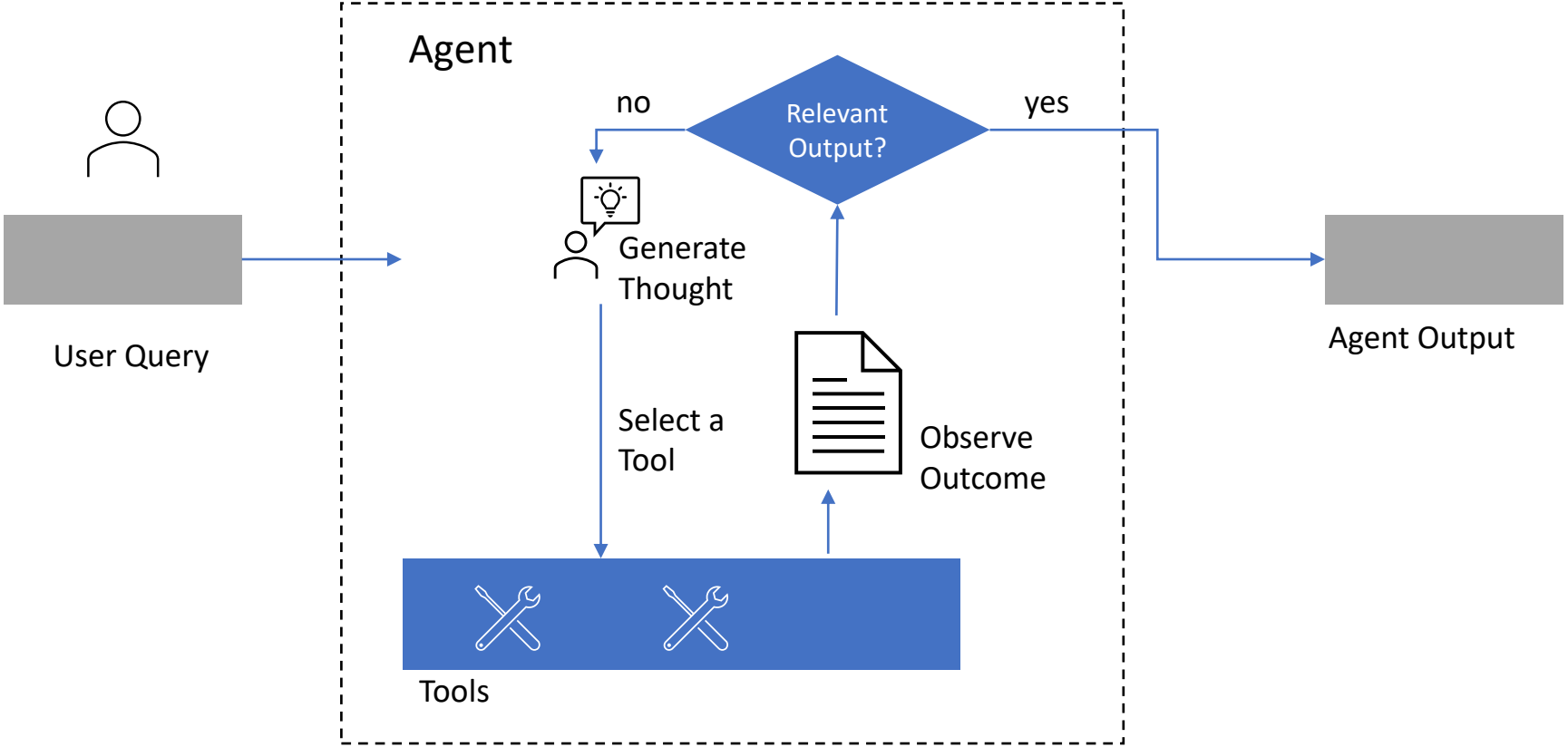
## What is an Agent?

- LLM apps execute tasks
- core element: agent
- uses planning, memory, and tools
- can perform actions
- is an expert in its field



# Agents

How does an Agent work?



# Agents

## Levels of AI Agents

Generality →

Performance ↑

| Level | Techniques | Performance | Capabilities | Key Characteristics | Use Cases | Narrow Domain | General Wide-Range Domain |
|---|---|---|---|---|---|---|---|
| 5 | LLM-based AI + Tools (Intent + Actions + Reasoning & Decision Making + Memory + Reflection + Autonomous Learning + Generalisation + Personality (Emotion + Character) + Collaborative behaviour (Multi-Agents) | Superhuman > 100% of Skilled Adults | True Digital Persona | Agent represents the user in completing affairs, interacts on behalf of user with others, ensuring safety & reliability. | Agent acts on behalf of user to complete tasks, interacting with others while ensuring safety & reliability. | Superhuman Narrow-AI AlphaFold, AlphaZero, StockFish | Artificial Super Intelligence (ASI) Not yet achieved |
| 4 | LLM-based AI + Tools (Intent + Actions + Reasoning & Decision Making + Memory & Reflection + Autonomous Learning + Generalisation | Virtuoso Equal to 99% of Skilled Adults | Memory & Context Awareness | Agent senses user context, understands user memory, and proactively provides personalised services at times. | A personalised virtual assistant enhances UX by understanding context & memory while acting proactively. | Virtuoso Narrow-AI AlphaGo, Deep Blue | Virtuoso AGI Not yet achieved |
| 3 | LLM-based AI + Tools (Intent + Actions) + Reasoning & Decision Making + Memory & Reflection | Expert Equal to 90% of Skilled Adults | Strategic task Automation | Using user-defined tasks, agents autonomously plan, execution steps using tools, iterates based on intermediate feedback until completion. | Agents autonomously plan and execute steps based on intermediate feedback | Expert Narrow-AI Purpose build, specific task orientated Agents | Expert AGI Not yet achieved |
| 2 | IL/RL-based AI + Tools (Intent + Actions) + Reasoning & Decision Making | Competent Equal to 50% of Skilled Adults | Deterministic Task Automation of Skilled Adults | Based on user description of deterministic task, agent auto-completes steps in predefine action. | User: "Check the weather in Beijing today". | Competent Narrow-AI Conversational AI build frameworks with LLM, RAG, etc. | Competent AGI Not yet achieved |
| 1 | Rule-Based AI + Tools (Intent + Actions) | Emerging Equal to Unskilled Humans | Simple Step Sequence | Agents complete tasks following exact steps, pre-defined by users or developers. | User: "Open Messenger" User: "Open the first unread email in my mailbox and read its content" User: "Call Alice". | Emerging Narrow-AI Single Rule-based systems, SHRDLU, GOFAI | Emerging AGI ChatGPT, Gemini, Llama 2, etc. |
| 0 | No AI Tools (Intent + Rules + Actions) | No AI | No AI | No AI | No AI | Narrow Non-AI UI Driven Software | General Non-AI Human-In-The-Loop Computing Mechanical Turk |

Adapted From: https://arxiv.org/pdf/2405.06643

Source: https://cobusgreyling.medium.com/5-levels-of-ai-agents-updated-0ddf8931a1c6

# Available Frameworks

# Agents

Which Frameworks are available?

## LangGraph

- built on top of LangChain (same team)
- flexible, customizable
- works with any LLM

- More complex

## CrewAI

- very intuitive
- suitable for many agents
- supports many LLM providers

- not ideal for very complex tasks

## Agents

- very easy to use
- suitable for beginners
- Works with any LLM

# Agents

Which Frameworks are available?

Microsoft

## AG2 (formerly: AutoGen)

- mostly for two agents
- good for code generation
- very powerful

## Google ADK

- Agent development kit
- Framework for development and deployment of AI agents
- Easy to learn
- Multi-agent architecture

# Agents

Which Frameworks are available?

Microsoft

## Magentic-One

- suitable for beginners
- pre-defined 5 agents: manager, web-surfer, file-surfer, coder, terminal
- built on top of AutoGen
- limited support and documentation

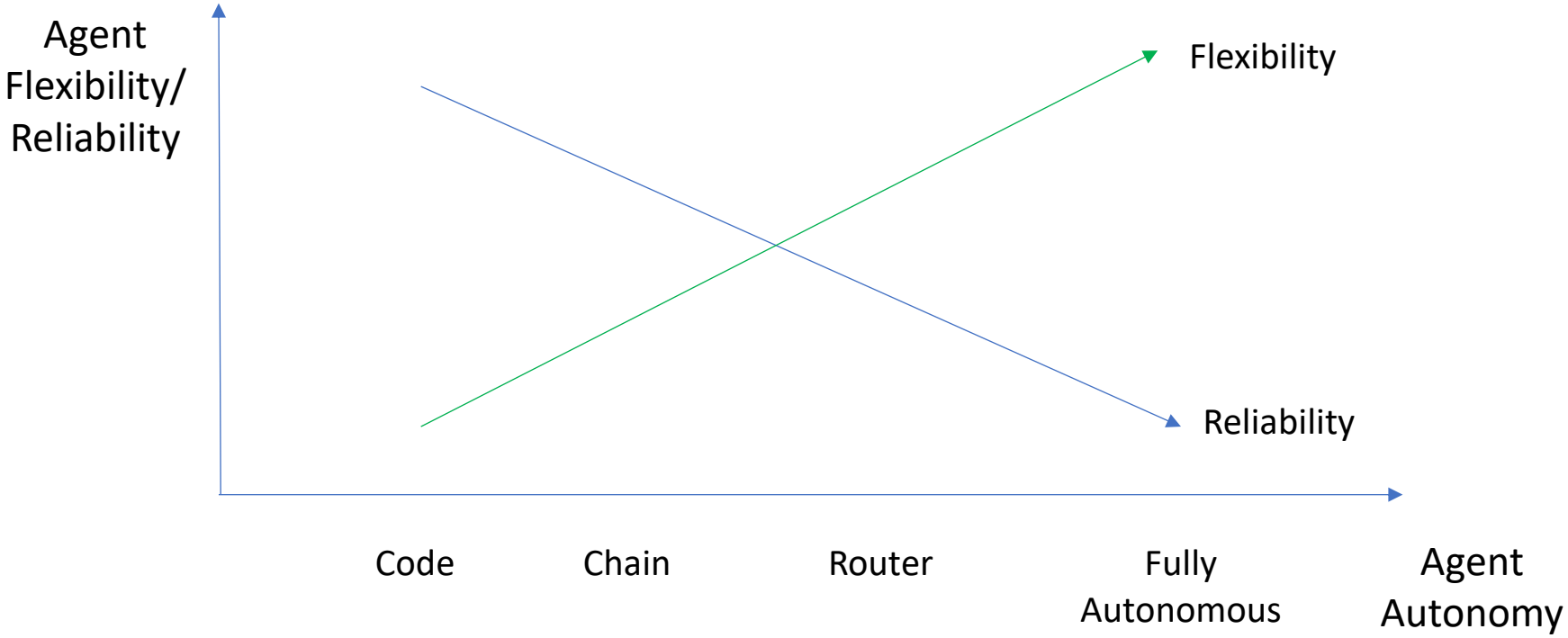
Microsoft

## tinytroupe

- multiagent persona simulation for imagination enhancement and business insights
- only works with GPT-4o
- Link

# Agents

## Flexibility / Reliability vs. Autonomy



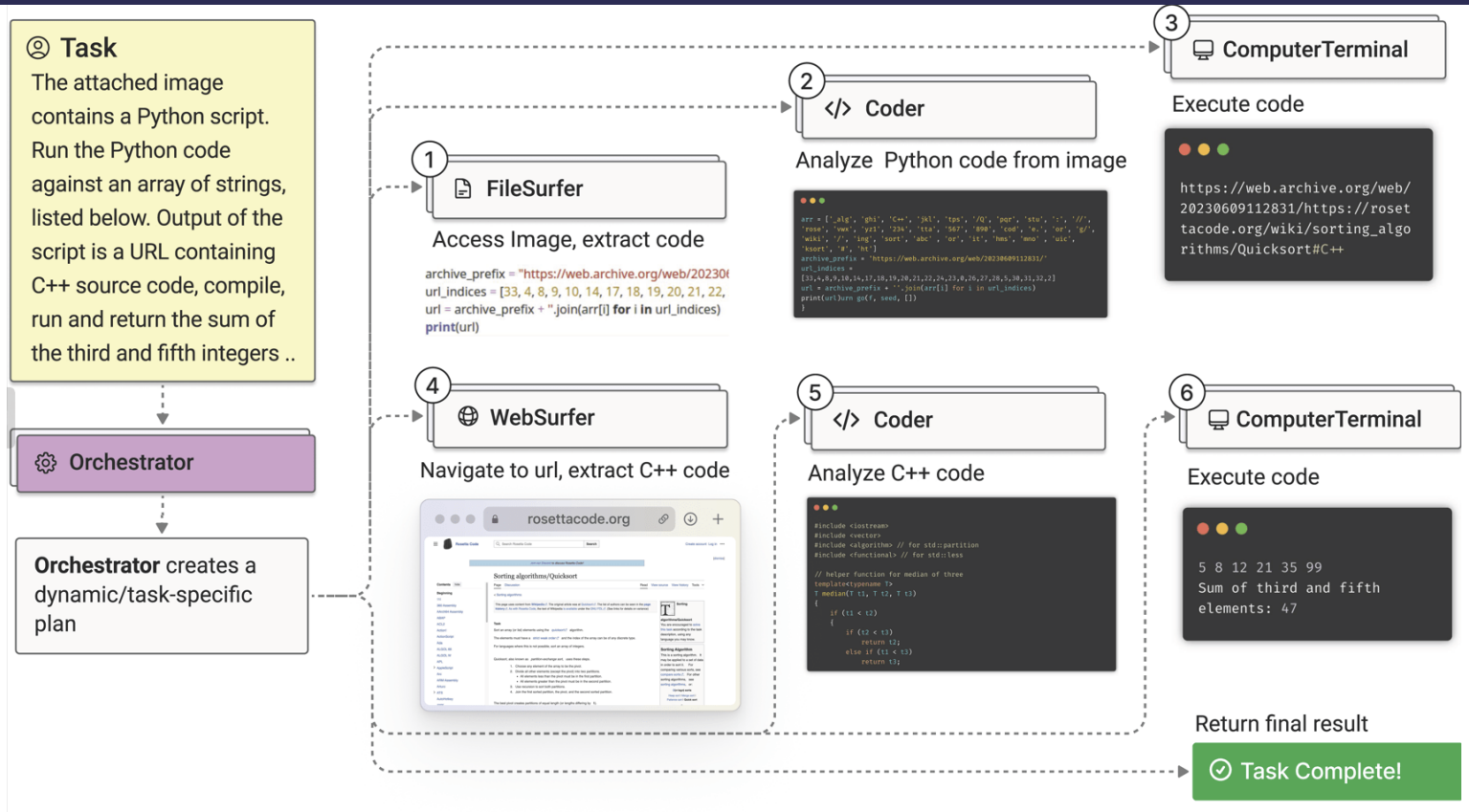

own graph; adapted from: LangChain Academy „Introduction to LangGraph“

# Agents

Flexibility vs.

# Agents

## Magentic One



# Agents

Tine Troupe

```python
factory = TinyPersonFactory("One of the largest banks in Brazil, full of bureaucracy and legacy systems.")

customer = factory.generate_person(
    """
    The vice-president of one product innovation. Has a degree in engineering and a MBA in finance.
    Is facing a lot of pressure from the board of directors to fight off the competition from the fintechs.
    """
)
```

✓ 10.1s Python

```python
customer.minibio()
```

✓ 0.0s Python

```
'Lucas Almeida is a 42 year old Vice-President of Product Innovation, Brazilian, currently living in Brazil.'
```

We can now perform the interview.

```python
customer.think("I am now talking to a business and technology consultant to help me with my professional problems.")
```

✓ 0.0s Python

```
Lucas Almeida --> Lucas Almeida: [THOUGHT]
                  > I am now talking to a business and technology consultant to help me with my
                  > professional problems.

TinyPerson(name='Lucas Almeida')
```

```python
customer.listen_and_act("What would you say are your main problems today? Please be as specific as possible.",
                        max_content_length=3000)
```

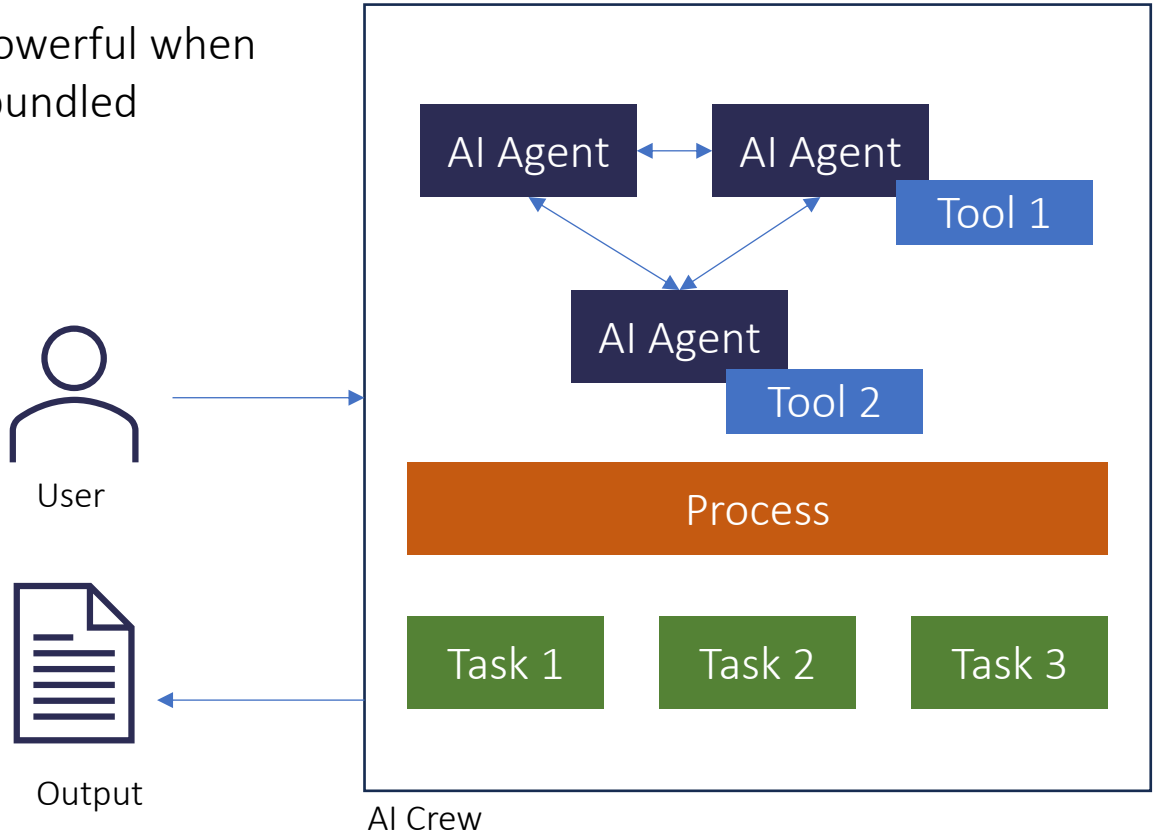
✓ 10.9s Python

# crewAI

# crewAI

## What is an AI Crew?

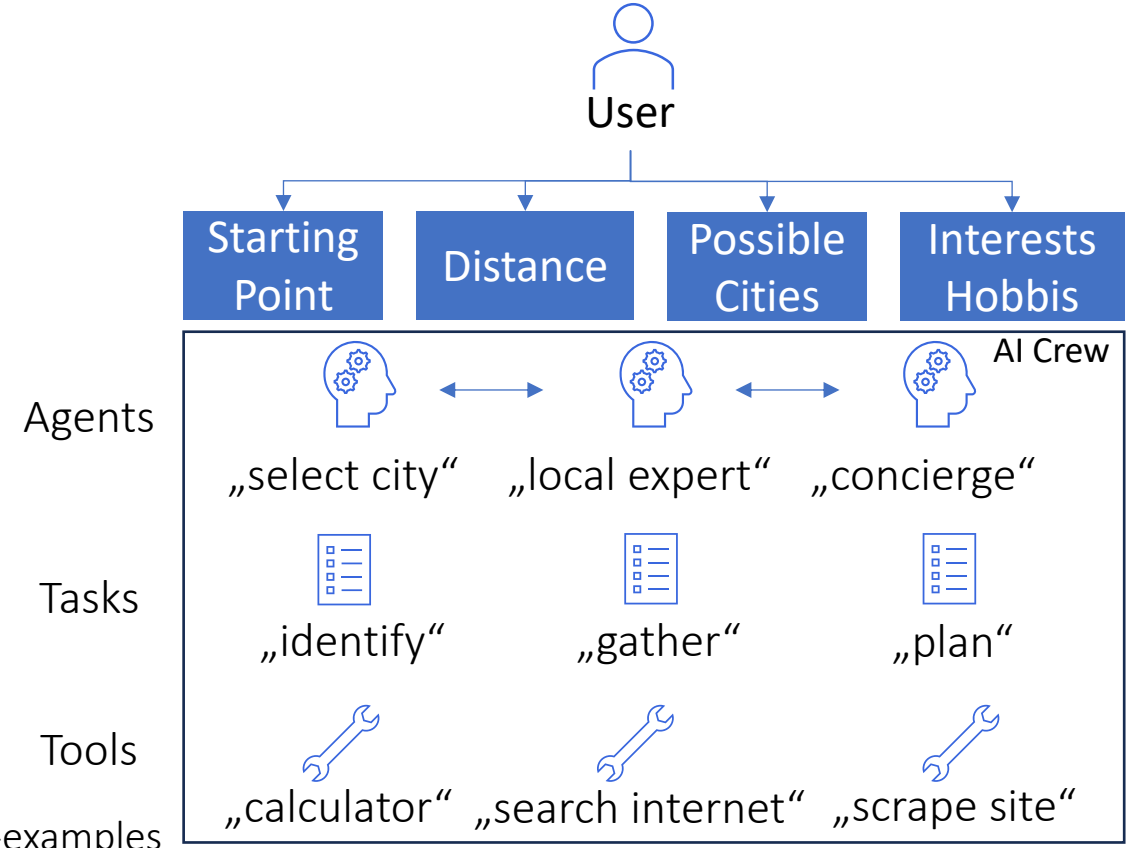
- concept extremely powerful when multiple agents are bundled
- system of experts



# crewAI

## Example – Plan your vacation

- 1. Define Goal
- 2. User Inputs
- 3. Set up
  - agents
  - tasks
  - if needed:
    - tools
    - process
    - …




Idea found at
https://github.com/joaomdmoura/crewAI-examples

# crewAI

## Example – Plan your vacation: Agents

AI Crew

Agents

| | „select city“ | „local expert“ | „concierge“ |
|---|---|---|---|
| Role | City Selection Expert | Local Expert | Travel Concierge |
| Goal | select city based on weather, season, prices | provide insights about selected city | Create travel itineraries with budget and packing suggestions |
| Backstory | You are an expert in analyzing travel data to pick the best destination. | You are a knowledgeable local guide with extensive knowledge about the city. | specialist in travel planning and logistics |

Tasks

„identify“ „gather“ „plan“

# crewAI

## Example – Plan your vacation: Tasks

AI Crew

Tasks

| | „identify“ | „gather“ | „plan“ |
|---|---|---|---|
| Description | … | … | … |

# crewAI

## File Dependencies

### agents.yaml

```yaml
researcher:
  role: >
    {topic} Data Researcher
  goal: >
    Find relevant news articles about
    {topic} for reputable sources.
  backstory: >
    You're a seasoned researcher with a
    knack for uncovering the latest
    developments in {topic}.
    Known for your ability to find the
    most relevant
    information and present it in a
    clear and concise manner.
```

### tasks.yaml

```yaml
information_gathering_task:
  description: >
    Conduct a thorough research about {top
    Make sure you find any interesting and
    the current year and month is {current
  expected_output: >
    A list with 10 bullet points of the mo
  agent: researcher
```

### crew.py

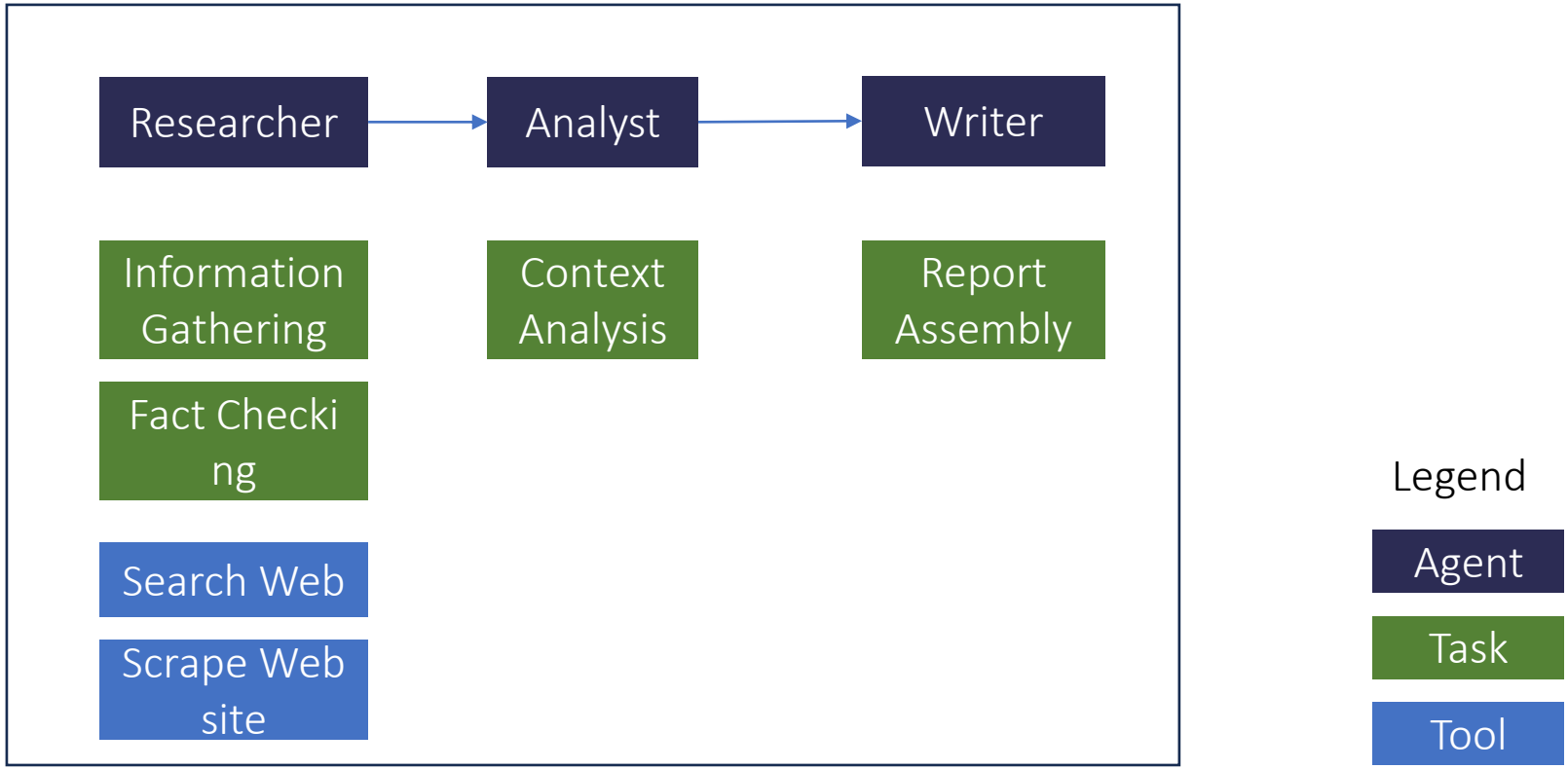
```python
@CrewBase
class NewsAnalysis():
    """NewsAnalysis crew"""

    tasks_config = 'config/tasks.yaml'
    agents_config = 'config/agents.yaml'

    @agent
    def researcher(self) -> Agent:
        return Agent(
            config=self.agents_config['researcher'],
            tools=[search_tool, website_search_tool],
            verbose=True
        )
```

### main.py

```python
def run():
    """
    Run the crew.
    """
    inputs = {
        'topic': 'AI Safety',
        'current_year_month':
        current_year_month
    }
    NewsAnalysis().crew().kickoff
    (inputs=inputs)
```

# crewAI

## Example – Crew Setup




AI Crew

# crewAI

## Tools

- can be used by Agents for

Searching the Internet

Scraping Websites

Reading Files

| Tool | Description |
|---|---|
| CodeDocsSearchTool | A RAG tool optimized for searching through code documentation and related technical documents. |
| CSVSearchTool | A RAG tool designed for searching within CSV files, tailored to handle structured data. |
| DirectorySearchTool | A RAG tool for searching within directories, useful for navigating through file systems. |
| DOCXSearchTool | A RAG tool aimed at searching within DOCX documents, ideal for processing Word files. |
| DirectoryReadTool | Facilitates reading and processing of directory structures and their contents. |
| FileReadTool | Enables reading and extracting data from files, supporting various file formats. |
| GithubSearchTool | A RAG tool for searching within GitHub repositories, useful for code and documentation search. |
| SerperDevTool | A specialized tool for development purposes, with specific functionalities under development. |
| TXTSearchTool | A RAG tool focused on searching within text (.txt) files, suitable for unstructured data. |

...

Source: https://docs.crewai.com/core-concepts/Tools/#available-crewai-tools

# crewAI

## Collaboration

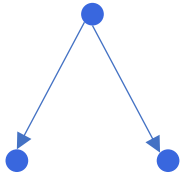
Agents can collaborate on a task to

- share information
- assist on a task
- allocate and optimize resources

```
from crewai import Agent, Task, Crew, Process

crew = Crew(
    agents=[planner, writer, editor],
    tasks=[plan, write, edit],
    verbose=2,
    manager_llm=llm,
    process= Process.hierarchical
)
```

Process.sequential

Process.hierarchical



# crewAI

## Expected Task Outcome

- output formats can be defined in detail

```
class OutputFormat(BaseModel):
    chapter_title: str
    bullet_points: list[str]
```

```
Task(
    description=(“...”),
    expected_output="A well-written slideset …“,
    agent=editor,
    output_format="markdown",
    output_format_model=OutputFormat,
    output_format_description=(
        "The output format is a markdown file …"
    ),
    output_file = "slideset.md"
)
```

# crewAI

## Use of other LLMs

- set up an llm-object
- pass it as a parameter

```
from langchain_groq import ChatGroq
```

```
llm=ChatGroq(temperature=0,
             model_name=MODEL,
             api_key=os.environ["GROQ_API_KEY"])
```

```
planner = Agent(
    role="…",
    goal="…",
    backstory="…",
    allow_delegation=False,
    llm=llm,
    verbose=True
)
```

# crewAI

## Installation

- Problem:
  - crewAI is a huge framework with many dependencies
  - Does currently have issues with Python 3.13

```
C:\Temp\crewai>where python
C:\Python313\python.exe
C:\Users\BertGollnick\AppData\Local\Programs\Python\Python312\python.exe
C:\Users\BertGollnick\AppData\Local\Microsoft\WindowsApps\python.exe
```

- Start from totally clean environment:

```
C:\Temp\crewai>C:\Users\BertGollnick\AppData\Local\Programs\Python\Python312\python.exe -m venv .venv
```

```
C:\Temp\crewai>.venv\Scripts\activate
```

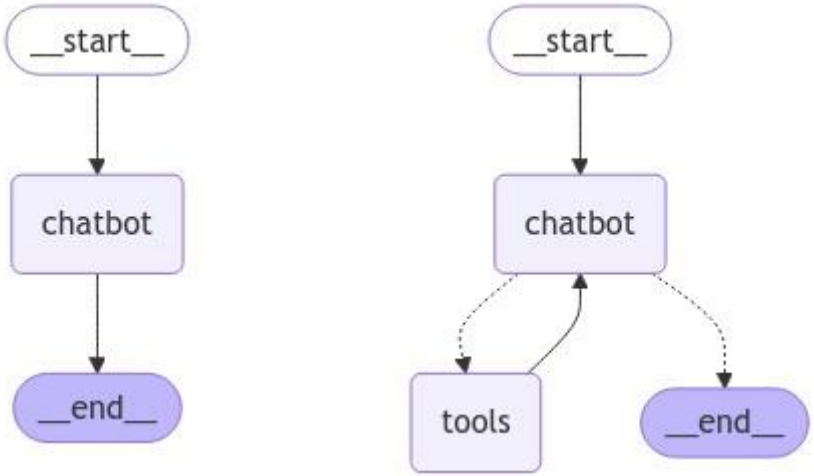
```
(.venv) C:\Temp\crewai>pip install crewai[tools]
```

# LangGraph

# LangGraph

Introduction

- agentic system
- graph-based representation (directed acyclical graph DAG)
- integrates well with LangChain ecosystem
- focuses on complex workflows
- based on nodes (tasks), and edges (dependencies)




Typical LangGraph graphs

# LangGraph

Chat with Summarization

messages:

System Message

cnt=2

Human Message

AI Message

messages:

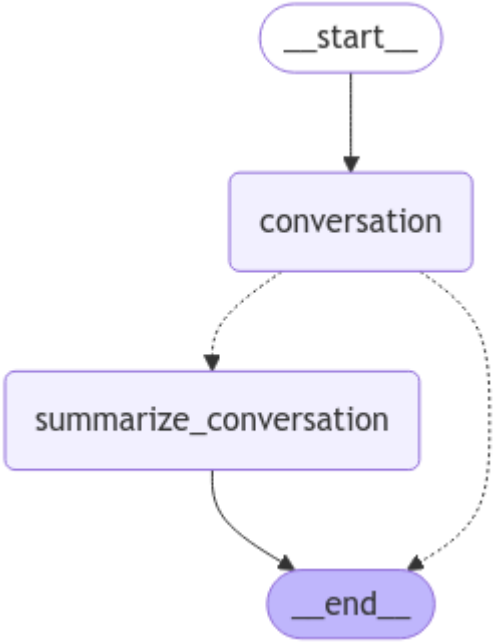
cnt>6:

Human Message

AI Message

summary:

Summary



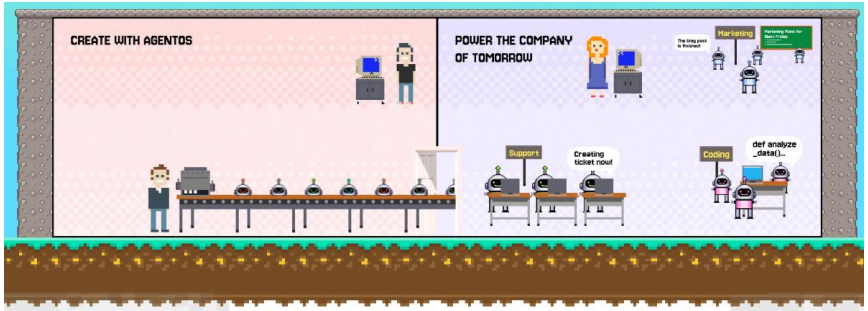
# AG2 (formerly autogen)

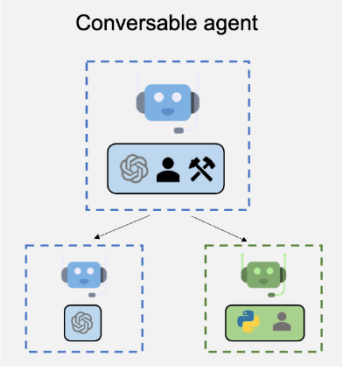
# AG2

## Introduction

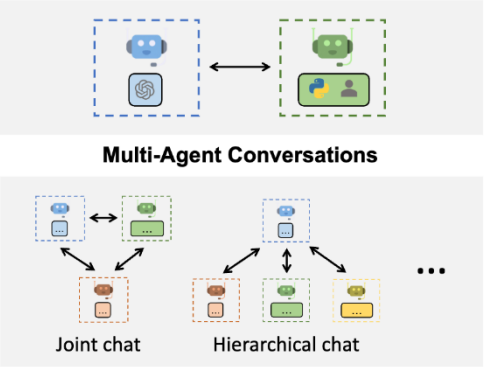
- Open-Source framework for building AI agents

- Installation: pip install ag2
- Docker: optional




Agent Customization

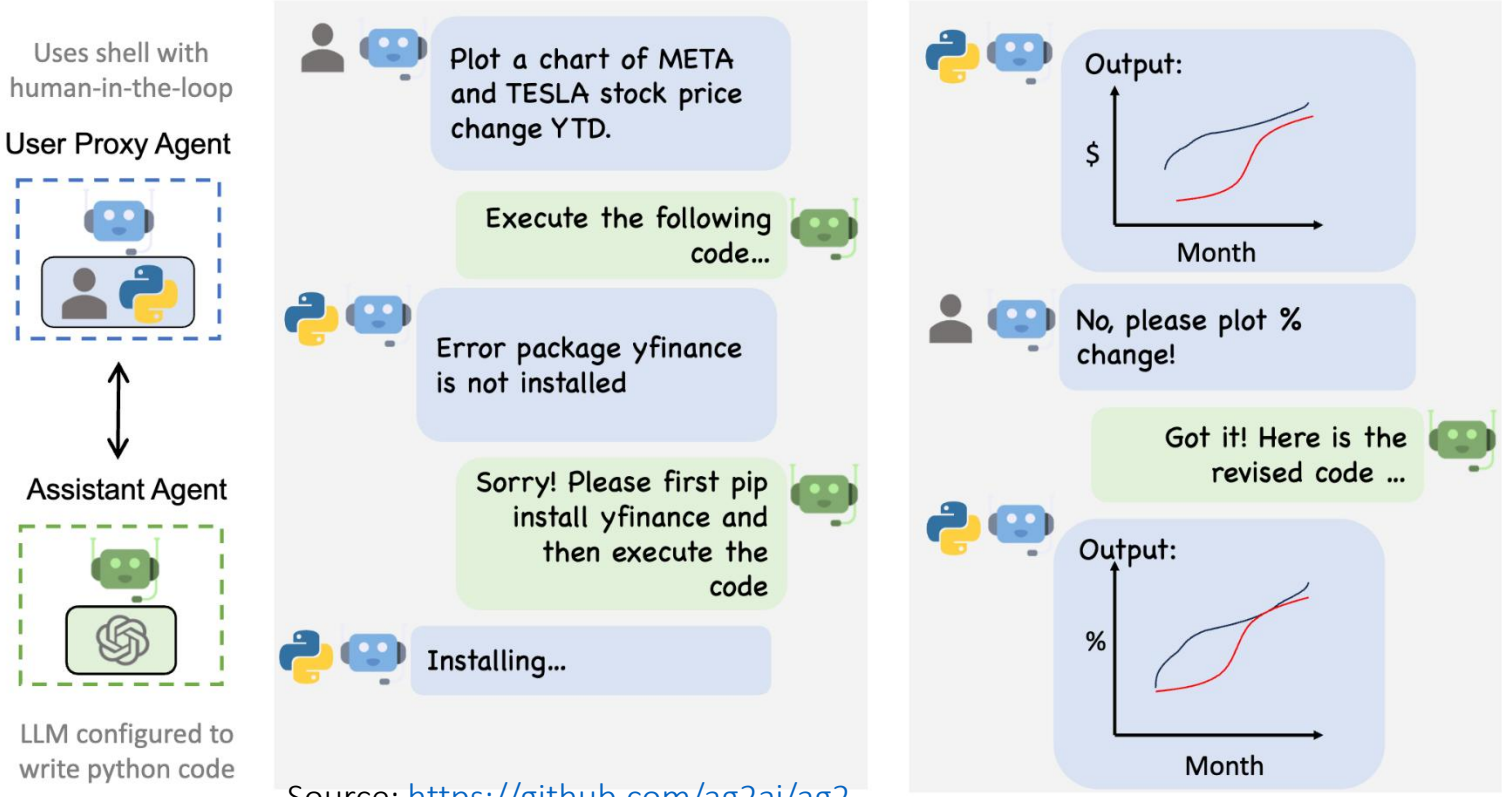



Flexible Conversation Patterns

Source: https://github.com/ag2ai/ag2

# AG2

## Introduction

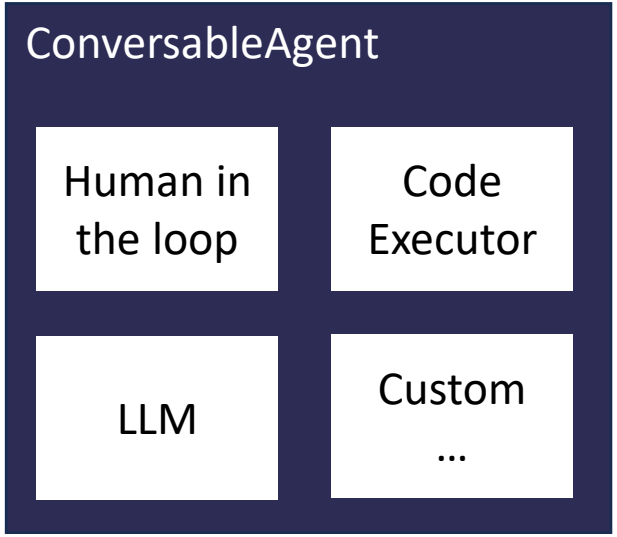



Source: https://github.com/ag2ai/ag2

# AG2

## Agent

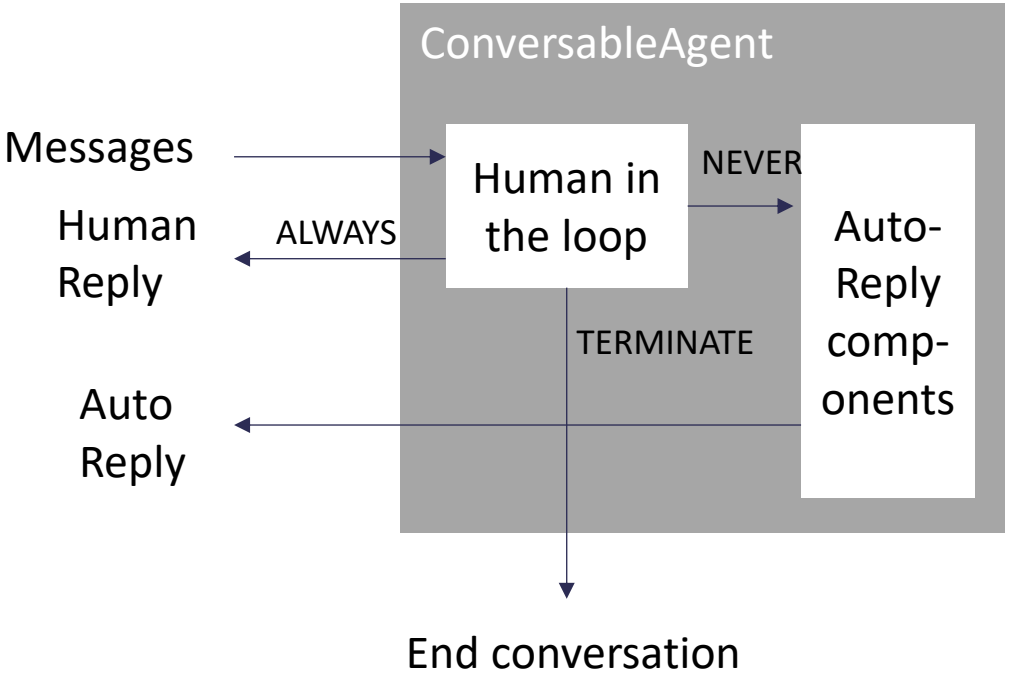
- AG2 agent
  - entity that can send and receive messages to and from other agents
  - agent can be run by
    - models,
    - code executors,
    - human, or
    - a combination of above



# AG2

## Human in the loop

- HITL in front of Auto reply, intercepts incoming messages, and decides to pass to auto-reply, or to human feedback
- customizable through *human_input_mode* parameter
- modes:
  - NEVER, TERMINATE (default), ALWAYS



# AG2

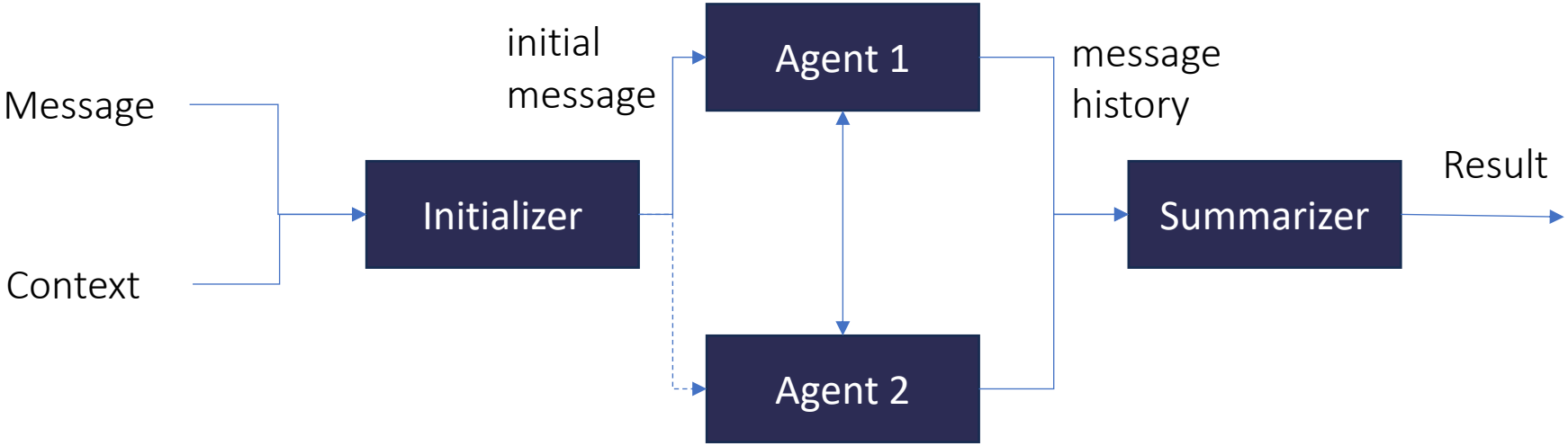
## Tools

- agents can use tools
- register_for_llm
  - exposes tool to LLM
  - allows LLM to reason about tool, decide when to call
- register_for_execution
  - handles execution of tool when LLM decides to call it
  - connects logical request generated by LLM to process
  - without it, even if LLM decides to use a tool, there would be no backend to execute tool‘s functionality

```
# %% create an agent with a tool
my_assistant = ConversableAgent(
    name="my_assistant",
    system_message="You are a helpful AI assistant.",
    llm_config=config_list
)

# register the tool signature at agent level
my_assistant.register_for_llm(
    name="get_current_date",
    description="Returns the current date in the form
at YYYY-MM-DD."
)(get_current_date)

# register the tool function at execution level
my_assistant.register_for_execution(name="get_current
_date")(get_current_date)
```

# AG2

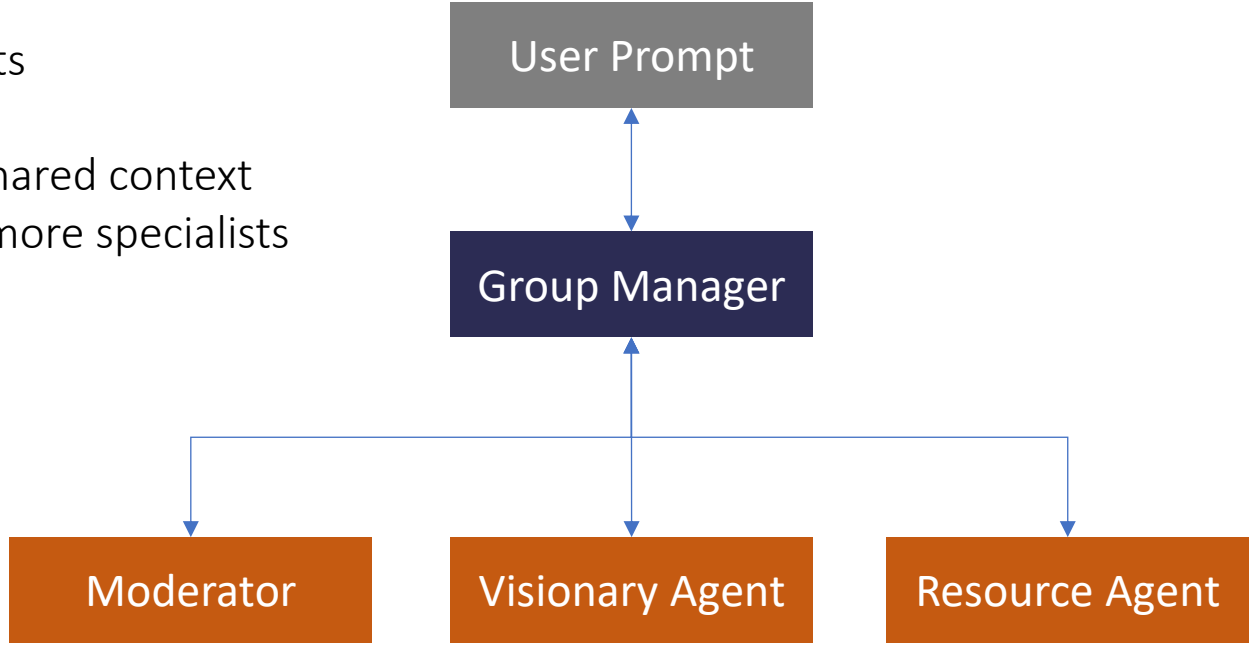
## Conversation Patterns: Two Agents Chatting



# AG2

## Group Chat

- Multiple agents interact
- Allows a team of specialists
- Dynamic collaboration
- All agents use the same shared context
- Easily scalable by adding more specialists



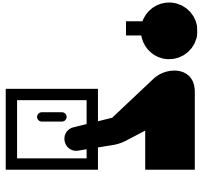
# OpenAI Agents SDK

# OpenAI Agents SDK

Introduction

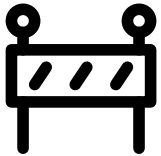
Agent

- LLM
- Instructions
- tools



Handoffs

- Agents delegate tasks to other agents

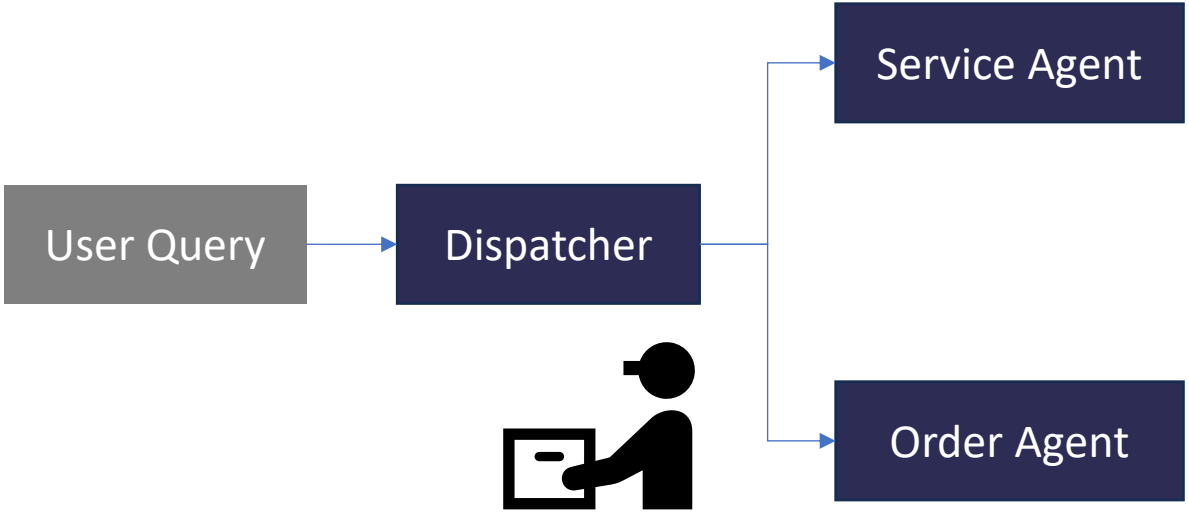


Guardrails

- Enables input validation

# OpenAI Agents SDK

## Handoff Implementation



# OpenAI Agents SDK

## Traces

### Traces

Workflow Search... | Group Search... | Metadata

| Workflow | Flow |
| --- | --- |
| Agent workflow | dispatcher ▸ service_agent |
| Agent workflow | dispatcher ▸ order_agent |

Traces / Agent workflow trace_b000faa931714b...

- dispatcher
  - POST /v1/responses
  - Handoff › service_agent
- service_agent
  - POST /v1/responses

**Span details**

**Properties**

| | |
| --- | --- |
| Created | Jul 7, 2025, 8:04 AM |
| ID | resp_686b6379e518819abf16... |
| Model | gpt-4o-2024-08-06 |
| Tokens | 114 total |
| Functions | `transfer_to_order_agent()` `transfer_to_service_agent()` |
| Configuration | |
| Response | text |

**Instructions**

System Instructions

```
You are a dispatcher. You determine which agent should handle the user's r
```

**Input** 101t

User

I ordered a while ago, but I haven't received my pizza yet. What's the status?

**Output** 13t

Function Call

# OpenAI Agents SDK

## Running Options

### Runner.run()

- Asynchronous
- Returns RunResult

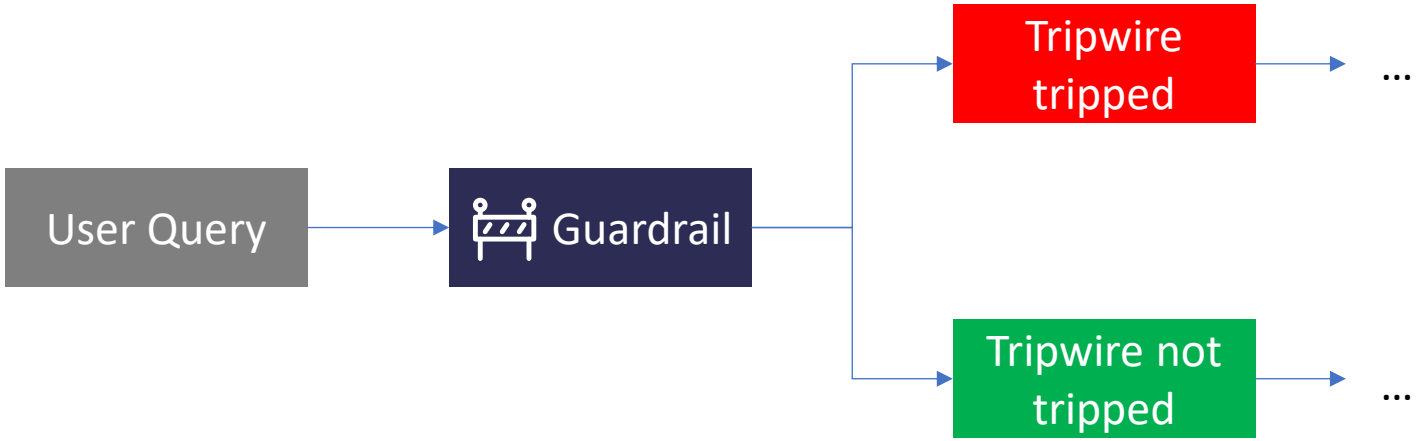
### Runner.run_sync()

- Synchronous
- Returns RunResult

### Runner.run_streamed()

- asynchronous
- Returns RunResultStreaming

# OpenAI Agents SDK

## Guardrails

- Run parallel to agents
- Allow for input validation